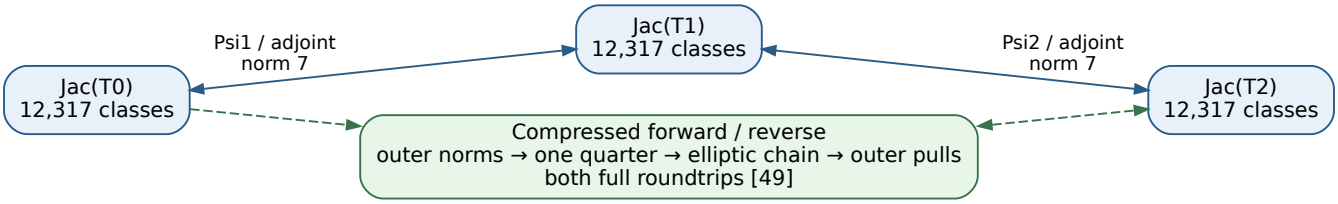
Actual prime-field second hop and compressed chain

2026-11-12 · fixed F101 models · scientific standing unsigned

The complete changed check passes:25 native tests,98 reference unit tests and five integrations,with one historical ignored benchmark. All13 source02 hashes stay unchanged. PCC3be24d preserves the entire155,019-byte raw log as standard XZ16,056 bytes,with raw/archive hashes,exact roundtrip and after-source receipt. The run started09:35:04UTC and ended10:22:08UTC;these intervals are correctness-run provenance,not performance measurements. [pcc:unverified]



What changed in the actual consumer

SuccessorChain.forward and dual now select cached standalone prime-field contexts. Exact previous bodies remain forward_reference and dual_reference; the new complete comparisons call them explicitly. Old generic,quarter, naive and original test bodies remain unchanged. Accepted T2 data,norm/pullback and forward/reverse evaluator bodies are reused after paths,visibility and reference-name adaptations. Existing field,polynomial,elliptic,Kohel and Balanced engines are reused. Source custody isd0ad0a. [pcc:unverified]

The standalone evaluators construct no Richelot or extension-field map. The retained comparison context still constructs those old maps for its reference,generic,naive and geometric tests. Redirecting the existing methods does not remove these cold construction costs. This is fixed public-fixture coverage,not a production catalog or general-prime implementation. [pcc:unverified]

New acceptance component	Complete retained controls
Two T2 norms	Each12,317 oldH2 outputs;group,model,poles,repeated support,infinities
Two T2 pullbacks	All109/113 oldR2P2 outputs;norm4,offdiagonal0;24,650 ordered sums
Second forward and adjoint	Each12,317 old outputs;both full norm7 compositions;actual beta/ betaHat
Compressed forward and reverse	Each12,317 old/staged outputs;both full norm49 compositions
Original elliptic sources	Every109/113 point/scalar,direct endpoint,reverse196,recovery104/64

All components retain identity,bijection,negation,fixed-partner addition, wrong-model and wrong-quarter controls. The original92 tests remain,including generic/naive,geometric kernels,branch automorphisms,independent7 certificates and chained transport. No case was filtered from the complete guard. [pcc:unverified]

Exact custody and failures

Thirteen source snapshots and before/after hashes bind this run. The initial missing reverse-child registration E0433/a749e8 remains with its full source and diagnostic;native tests did not run on that failed version. A custody comparison first mistook rustfmt wrapping after reference renaming for a source change;the failed driver/status are retained,and the accepted guard formats the reference-name adaptation before exact comparison. Arithmetic and acceptance criteria stayed unchanged. [pcc:unverified]

Global formulas,exceptions and recovery

SECOND_CHAIN_PROOF.md retains the admitted Picard/quotient/polarization dependencies. Write K_i for the old product-to-Jacobian pulls and H_i for their adjoint norms. H_iK_i and K_iH_i are[4]. The old maps satisfy 4Psi1=K1 alpha H0 and4Psi2=K2 beta H1. The middle relation gives 16Psi2Psi1=4K2 beta alpha H0;cancel4 in Hom to obtain4F=K2 beta alpha H0. Taking adjoints gives4FHat=K0 alphaHat betaHat H2. Point groups are not assumed torsion-free. [pcc:unverified]

On rational groups of order12,317=109 times113,the single public quarter is82 modulo109 and85 modulo113. Compressed forward takes T0 norms,alpha then beta,and T2 pulls;reverse takes T2 norms,betaHat then alphaHat,and T0 pulls. No middle T1 class occurs in these expressions. The composite retains geometric degree2,401 and both adjoint norms49. [pcc:unverified]

T2 component	N,Q,M coefficient vectors,low first	Infinity image
(59,27),order109	[97,93,7,0,97];[55,0,1];[71,49,20,63]	(97,63)
(23,7),order113	[74,0,81,42,64];[23,0,1];[37,9,9,35]	(64,35)

Exact curve identities and the old norm4/degree argument identify these quartics globally using a difference-degree bound16. Independent flat replay checks finite points,infinities and every geometric pole. The first T2 pole orbit has degree4 and is absent from F101^6;it creates no degree≤2 rational pole support. The second has rational poles(49,12)/(49,89). Native exception counts keep both outcomes. Finite-fibre units and jets handle repeated supports;the signed infinity degree drops are[2,1]. [pcc:unverified]

Reverse normalization and public recovery

BetaHat uses cubics[13,62,81,1]/[11,64,23,1] and alphaHat uses [58,26,58,1]/[29,88,45,1]. Each normalized Kohel dual needs checked scale29 before the next stage. Since49 times4776 equals1+19 times12317,the rational inverse is[4776]FHat. For an original embedded elliptic source,H0K0 adds4, so the returned source norm is[196],recovered with104 modulo109 or64 modulo113. These are public scalars,not logarithm labels. [pcc:unverified]

Faithful upstream Lean modules cover conditional norm49/source196,two-hop clearing and rational recovery. Actual varieties,Picard maps,projective extension,global kernel degree and geometric distinction remain written or partial. Full rational replay does not prove those geometric properties. Independent whole-map/kernel replay is separate upstream work. [pcc:unverified]

Integration and next checks

This child starts immediate ready parent1507 at93587b9,tree0f032e. Parent exact CI43191178913 succeeds25native/92reference+five;whole225,939-byte log to XZ29,088/API/hash/roundtrip isf72285/root5bbe4d. This98-test head needs its own exact-head CI and review. Integrate bottom-up,retarget/rerun after parents land,and respect the frozen-ledger forks and actual merge gates. [pcc:unverified]

Global sources are crypto b351d685:T2_QUARTIC_PROOF/f9cf19, CHAIN_COMPRESSION_PROOF_V2/89e13d,REVERSE_CHAIN_PROOF/001dff and old Picard/quotient proofs. Old proofs,reports and negative controls remain frozen. Cold construction/transfer/solve/recovery/verification costs await actual S3/live-budget admission9e1812;no paid run,solver,novelty or measured advantage is claimed. Atlas vault remains locked and KB/ecdb unavailable. [pcc:unverified]